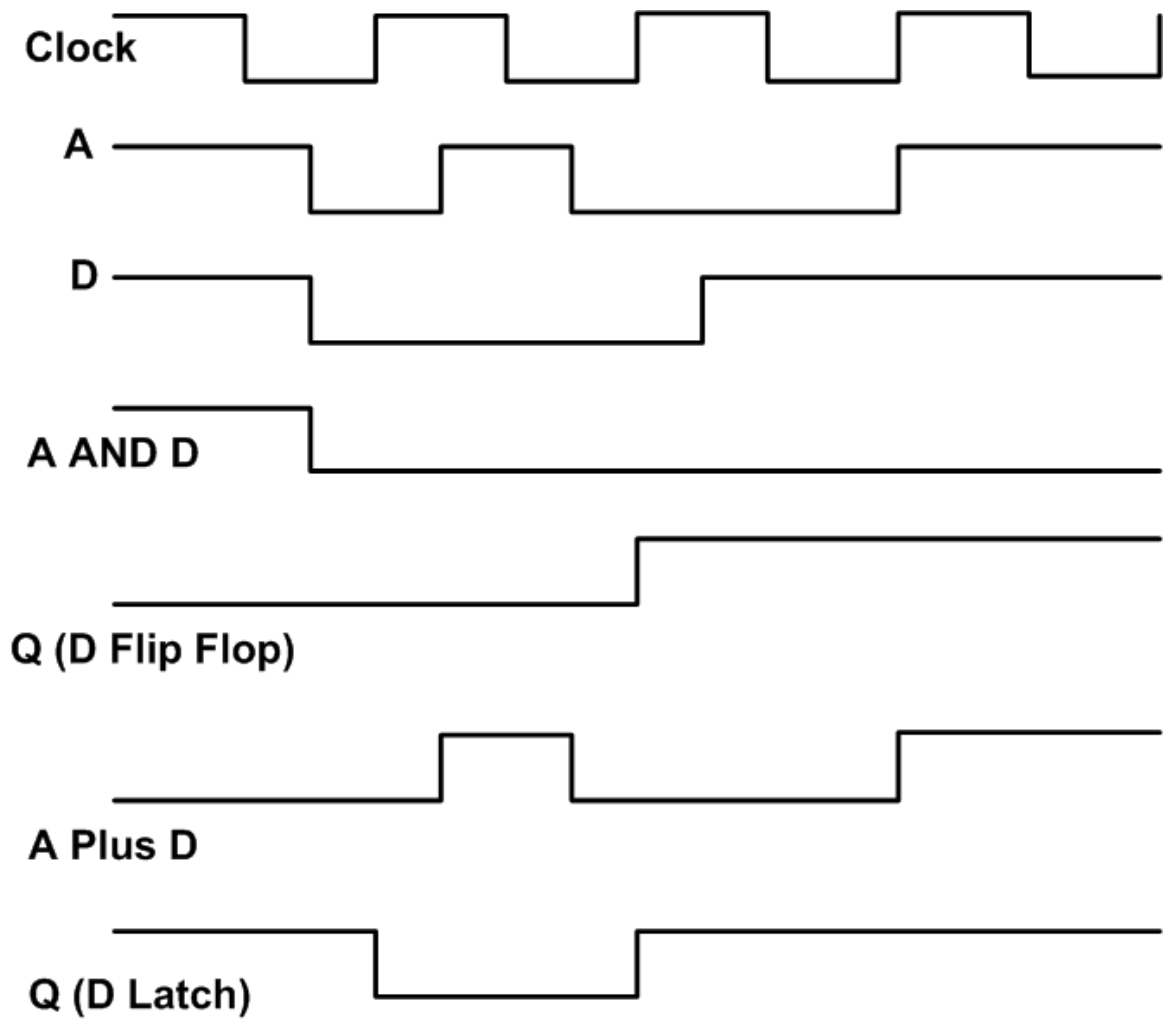


EC 551 Advanced Digital Design with Verilog and FPGAs

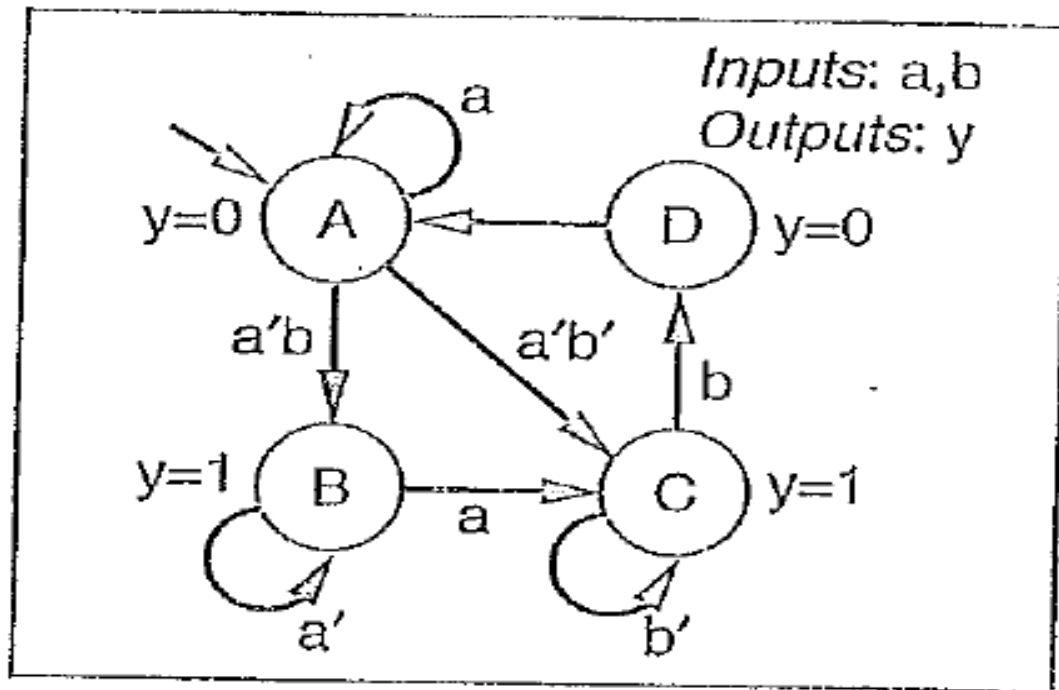
Fall 2026, Prof. Douglas Densmore

Sequential Logic Take Home Quiz

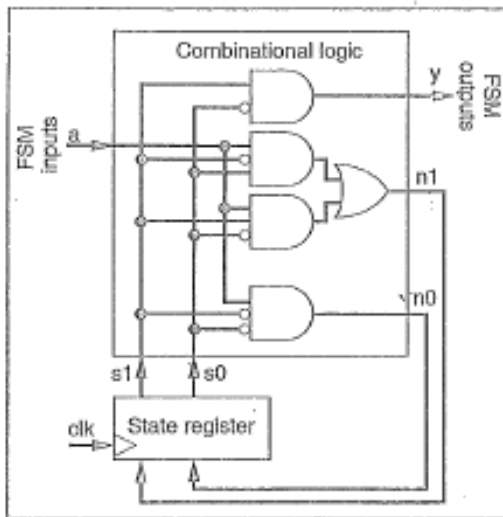
1. Circle five mistakes on this timing diagram. Explain very briefly why they are mistakes.



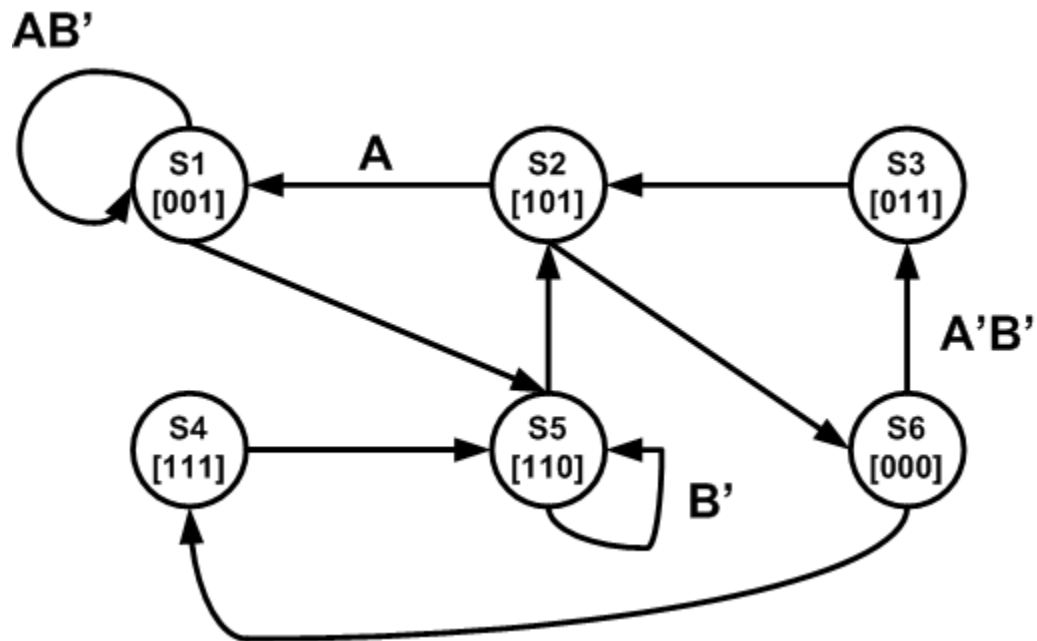
2. Create the truth table and Boolean equations for the next state functions and output function for the given FSM diagram. Create the controller design for this FSM consisting of inputs, outputs, current state register, and combinational logic for the next state and output functions.



3. Reverse engineer the FSM behavior of the sequential circuit provided.



4. Label the following FSM transitions which are missing information such that the machine has both exclusive and complete transitions. The transition conditions should be minimal.



5. Show that the final set of transitions (those you added plus what was there) is complete and exclusive for state S1.

6. How many flip flops will be needed for this FSM given a minimum bitwidth encoding? How many with a one-hot encoding? Be sure to explain your reasoning. Can you use output encoding (why or why not)?

7. Does the following state table represent a Mealy or Moore FSM? Why?

Inputs		Outputs		
X	Current State	Next State	z	clear
0	0	1	0	1
1	0	1	0	1
0	1	1	0	0
1	1	0	1	0

8. Draw the FSM diagram for this state table. Label it according to its FSM type.

9. Label the provided FSM (you can potentially label all arcs and all state output values if appropriate). This FSM should either a Mealy or Moore machine to reflect the opposite of what you determined “parts 7 and 8” were.

